

Uber Booking Analysis

1. INTRODUCTION

The **Uber Booking Analysis** project focuses on analyzing ride booking data to understand booking patterns, vehicle performance, customer behavior, payment methods, ride completion, and cancellation trends.

SQL was used for data analysis and querying, while Power BI was used to create an interactive dashboard with DAX measures and visualizations.

2. PROJECT OBJECTIVE

- Analyze the overall booking volume.
- Identify the most frequently used vehicle types.
- Analyze completed, cancelled, and incomplete bookings.
- Identify the most popular pickup and drop locations.
- Analyze customer and driver cancellation reasons.
- Analyze payment method usage.
- Calculate ride distance and customer rating metrics.

- Build an interactive Power BI dashboard for business insights.

3. Tools and Technologies

- **SQL / MySQL** – Data analysis and querying
- **Power BI** – Dashboard development
- **DAX** – Measures and dynamic calculations
- **Excel / CSV** – Dataset source and initial data handling

4. Data Preparation

The dataset was examined and cleaned before analysis.

Major data preparation activities included:

- Identifying missing values.
- Removing duplicate records where required.
- Checking incorrect or inconsistent values.
- Verifying column data types.
- Preparing the cleaned dataset for SQL and Power BI analysis.

5. SQL Analysis

SQL was used to answer business questions from the Uber booking dataset.

Key analysis areas included:

- Total number of bookings.
- Booking status analysis.
- Vehicle type-wise bookings.
- Payment method analysis.
- Top pickup locations.
- Top drop locations.
- Average ride distance.
- Customer cancellation reasons.
- Driver cancellation reasons.
- Completed booking percentage.

6. Power BI Dashboard

An interactive **Uber Booking Analysis Dashboard** was developed using Power BI.

KPI Cards

The dashboard contains important KPIs such as:

- Total Bookings
- Completed Rides
- Completion Percentage
- Average Ride Distance
- Average Customer Rating

Dashboard Visualizations

The dashboard includes:

- Booking Status Donut Chart
- Vehicle Type Analysis
- Payment Method Analysis
- Top 10 Pickup Locations
- Top 10 Drop Locations
- Average Customer Rating by Vehicle Type
- Customer Cancellation Reasons
- Driver Cancellation Reasons

Dashboard Filters

Interactive slicers were included for:

- Date
- Vehicle Type
- Booking Status
- Payment Method

7. DAX Measures

DAX measures were created to perform dynamic calculations in Power BI.

Total Bookings

```
Total Bookings =  
COUNTROWS(uber_bookings)
```

Completed Rides

```
Completed Rides =  
CALCULATE(  
    COUNTROWS(uber_bookings),  
    uber_bookings[Booking_Status] = "Completed"  
)
```

Completion Percentage

```
Completion % =  
DIVIDE(  
    [Completed Rides],  
    [Total Bookings],  
    0  
)
```

These measures allow the dashboard values to dynamically change when users apply filters and slicers.

8. Business Questions

The project was designed around practical business questions such as:

1. What is the total number of bookings?
2. How many rides were completed, cancelled, and incomplete?
3. Which vehicle type has the highest number of bookings?
4. Which payment method is used the most?
5. What are the top 10 pickup locations?
6. What are the top 10 drop locations?
7. Which vehicle type has the highest average ride distance?
8. What are the major customer cancellation reasons?
9. What percentage of bookings were completed successfully?
10. Which customers have the highest number of bookings?

9. Key Insights

The dashboard helps business users understand:

- Overall booking performance.
- The most popular vehicle categories.
- Successful versus cancelled bookings.
- High-demand pickup and drop locations.
- Common reasons for customer and driver cancellations.

- Customer rating patterns.
- Payment preferences.
- Ride distance patterns.

Note: Exact numerical findings should be added from the final Power BI dashboard rather than using assumed values.

10. Project Outcome

The final outcome is an interactive Power BI dashboard that allows users to explore Uber booking performance through KPIs, charts, and interactive filters.

This project demonstrates an end-to-end **Data Analyst workflow**:

Raw Data → Data Cleaning → SQL Analysis → DAX Measures → Power BI Dashboard → Business Insights

11. Conclusion

The Uber Booking Analysis project demonstrates how SQL and Power BI can be combined to transform raw booking data into meaningful business insights.

SQL was used to analyze and prepare data, while Power BI and DAX were used to create an interactive dashboard. The project provides a practical understanding of data cleaning, SQL querying, business analysis, DAX measures, visualization, and dashboard development.

12.PROJECT DASHBOARD

Uber



UBER BOOKING ANALYSIS DASHBOARD

150K
Total Bookings



24.48
Avg Ride Distance/km



4.59
Avg Customer Rating



33.61K
Avg Booking Value



Vehicle Type

All

Booking Status

All

Payment Method

All

Date

1/1/2024

12/30/2024

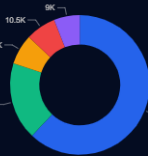
Ratings

4.23

4.30

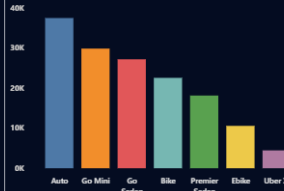
4.40

Booking Status

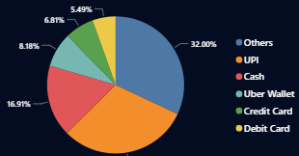


- Completed
- Cancelled by Driver
- Cancelled by Cust...
- No Driver Found
- Incomplete

Vehicle Type Analysis



Payment Method

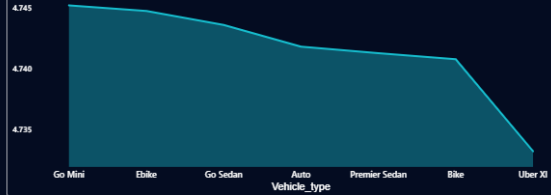


- Others
- UPI
- Cash
- Uber Wallet
- Credit Card
- Debit Card

Pickup Location



Avg Customer Rating



Drop Location

Ashok Park Main	861
Akshardham	856
Ashok Vihar	851
AIIMS	850
Adarsh Nagar	845
Ambience Mall	836
Anand Vihar	830

Canceling By Customer

